

Tao 9

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1 Continuous Functions

Let $x_0 \in \overline{E}$. $f(x) \rightarrow L$ as $x \rightarrow x_0$ in E iff for all sequences (x_n) in E such that $x_n \rightarrow x_0$, we have $f(x_n) \rightarrow L$.

Suppose $f(x) \rightarrow L$ as $x \rightarrow x_0$ in E . Let (x_n) in E be such that $x_n \rightarrow x_0$. Let $\epsilon > 0$. By definition of convergence of f , pick $\delta > 0$ such that $x \in V_\delta(x_0) \cap E \rightarrow f(x) \in V_\epsilon(L)$. By definition of convergence of (x_n) , pick N such that $n \geq N \rightarrow x_n \in V_\delta(x_0)$. Since $x_n \in E$ by construction, we can combine to get $x_n \in V_\delta(x_0) \cap E$, so $f(x_n) \in V_\epsilon(L)$. Hence, $f(x_n) \rightarrow L$.

Conversely, suppose there exists $\epsilon > 0$ such that for all $\delta > 0$, we have $x \in V_\delta(x_0) \not\rightarrow f(x) \in V_\epsilon(L)$. Take $\delta = 1/n$, so there exists $x_n \in V_\delta(x_0)$ for which $f(x_n) \notin V_\epsilon(L)$. Since $x_n \rightarrow x_0$ but $f(x_n) \not\rightarrow L$, we have a contradiction.

Let $f : X \rightarrow \mathbb{R}$ be a function, $E \subset X$ and $x_0 \in \overline{E}$. Define $\limsup_{x \rightarrow x_0; x \in E} f(x) = \inf_{\delta > 0} \sup f(V_\delta(x_0) \cap E)$. I claim that $\limsup$ is local: for all $\delta_0 > 0$, $\limsup_{x \rightarrow x_0; x \in E} f(x) = \limsup_{x \rightarrow x_0; x \in E \cap V_{\delta_0}(x_0)} f(x)$.

Proof: Let $A = \{\sup f(V_\delta(x_0) \cap E) : \delta > 0\}$ and $B = \{\sup f(V_\delta(x_0) \cap V_{\delta_0}(x_0) \cap E) : \delta > 0\} = \{\sup f(V_\delta(x_0) \cap E) : 0 < \delta \leq \delta_0\}$, so $B \subset A$. It follows $\inf A \leq \inf B$. Further, since $V_\delta(x_0) \cap E \supset V_\delta(x_0) \cap E \cap V_{\delta_0}(x_0)$, we have $\sup f(V_\delta(x_0) \cap E) \geq \sup f(V_\delta(x_0) \cap E \cap V_{\delta_0}(x_0))$, so for all $x \in A$, there exists $y \in B$ such that $x \geq y$. It follows $\inf A \geq \inf B$, so $\inf A = \inf B$.

Suppose $\lim_{x \rightarrow x_0; x \in E} f(x) = L$ exists. Take $\epsilon = 1$ and pick $\delta_0 > 0$ such that $f(V_{\delta_0}(x_0) \cap E) \subset V_1(L)$. Then $f|_{V_{\delta_0}(x_0) \cap E}$ is bounded. Since $\limsup_{x \rightarrow x_0; x \in E} f(x) = \limsup_{x \rightarrow x_0; x \in E \cap V_{\delta_0}(x_0)} f(x)$, we may assume that $f|_E$ is bounded.

Let $\epsilon > 0$. Pick $\delta > 0$ such that $f(V_\delta(x_0) \cap E) \subset V_\epsilon(L)$. Then $\limsup_{x \rightarrow x_0; x \in E} f(x) \leq L + \epsilon$. Since this holds for all $\epsilon > 0$, we have $\limsup_{x \rightarrow x_0; x \in E} f(x) \leq L$. Similarly, $\liminf_{x \rightarrow x_0; x \in E} f(x) \geq L$. Since in general $\liminf_{x \rightarrow x_0; x \in E} f(x) \leq \limsup_{x \rightarrow x_0; x \in E} f(x)$, we have that

$$\liminf_{x \rightarrow x_0; x \in E} f(x) = \limsup_{x \rightarrow x_0; x \in E} f(x) = L.$$

2 Real Exponentiation

Let $x \in \mathbb{R}^+$ and $n \in \mathbb{Z}^+$. Define $E = \{y \in \mathbb{R} : y^n \leq x\}$. Since $0^n = 0 \leq x$, $0 \in E$, so $E \neq \emptyset$. Pick $N \in \mathbb{Z}^+$ such that $x \leq N$ by the Archimedean principle. If $y \in E$ such that $y < 0$, we're done. Else, $y > 0$. Then $y^n \leq x \leq N \leq N^n$. If $y > N$, then $y^n > N^n$, absurd. Hence, $y \leq N$, so E is bounded above by N . By Completeness of $\mathbb{R}$, let $\alpha = \sup E$.

By Archimedean principle, pick $m \in \mathbb{Z}^+$ such that $\frac{1}{m} < x$. Then $\frac{1}{m^n} \leq \frac{1}{m} < x$, so $\frac{1}{m} \in E$, so $\alpha \geq \frac{1}{m} > 0$.

Assume FTSOC $\alpha^n > x$. Pick $0 < \epsilon < \min(\alpha, \frac{\alpha}{n} \cdot (1 - \frac{x}{\alpha^n}))$. Then

$$\begin{aligned} (\alpha - \epsilon)^n &= \alpha^n \cdot \left(1 - \frac{\epsilon}{\alpha}\right)^n \geq \alpha^n \cdot (1 - n\epsilon/\alpha) \\ &> \alpha^n \cdot \left(1 - \frac{n}{\alpha} \cdot \frac{\alpha}{n} \cdot \left(1 - \frac{x}{\alpha^n}\right)\right) = x. \end{aligned}$$

On the other hand, since $\alpha - \epsilon < \alpha$, we have $\alpha - \epsilon < y$ for some $y \in E$. Hence, $(\alpha - \epsilon)^n < y^n \leq x$, contradicting $(\alpha - x)^n > x$ from before.

Now suppose FTSOC $\alpha^n < x$, so $\frac{1}{x} < \frac{1}{\alpha^n}$. Pick $\epsilon > 0$ such that

$$\left(\frac{1}{\alpha} - \frac{\epsilon}{\alpha}\right)^n = \frac{1}{\alpha^n} \cdot (1 - \epsilon)^n \geq \frac{1 - n\epsilon}{\alpha^n} > \frac{1}{x},$$

that is, when $0 < \epsilon < \min(1, \frac{1}{n} \cdot (1 - \frac{\alpha^n}{x}))$. Rewrite $\frac{1}{\alpha} - \frac{\epsilon}{\alpha} = \frac{1}{\alpha + \delta}$ for some $\delta > 0$. Then $(\alpha + \delta)^n < x$, so $\alpha + \delta \in E$, contradicting maximality of α .

By Trichotomy of $\mathbb{R}$, we have $\alpha^n = x$, showing existence of n th roots.

Suppose $\alpha, \alpha' \in \mathbb{R}^+$ such that $\alpha^n = (\alpha')^n = x$. If $\alpha < \alpha'$, then $\alpha^n < (\alpha')^n$. If $\alpha > \alpha'$, then $\alpha^n > (\alpha')^n$. Hence, $\alpha = \alpha'$, showing uniqueness of n th roots. Symbolically write $\alpha = x^{1/n}$.

Then for $x, y \in \mathbb{R}^+$, $(x^{1/n}y^{1/n})^n = (x^{1/n})^n(y^{1/n})^n = xy$, so $x^{1/n}y^{1/n} = (xy)^{1/n}$, since $x^{1/n}y^{1/n} > 0$.

Let $x \in \mathbb{R}^+$, $m, n \in \mathbb{Z}^+$. Then $x^{1/n} \in \mathbb{R}^+$, so we can consider $(x^{1/n})^{1/m} \in \mathbb{R}^+$. In particular

$$\left((x^{1/n})^{1/m}\right)^{m \cdot n} = \left(\left((x^{1/n})^{1/m}\right)^m\right)^n = (x^{1/n})^n = x,$$

so $x^{1/(m \cdot n)} = (x^{1/n})^{1/m} = (x^{1/m})^{1/n}$, since multiplication integers is commutative.

Let $x \in \mathbb{R}^+$. Then $x^n \in \mathbb{R}^+$. Trivially, $(x^n)^{1/n} = x$.

Now define formally for $a, b \in \mathbb{Z}^+$ the value $x^{a/b} = (x^{1/b})^a$.

3 Continuity

Let $a > 0$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by $f(x) = a^x$. Then f is continuous.

Proof: Suppose $(x_n) \subset \mathbb{R}^\omega$ converges to x . For each $n \in \mathbb{Z}^+$, pick a sequence of rationals (x_{nm}) such that $x_{nm} \rightarrow x_n$. Since $(a^{x_{nm}}) \rightarrow a^{x_n}$, pick $M \in \mathbb{Z}^+$ such that $m \geq M \rightarrow |a^{x_{nm}} - a^{x_n}| < 1/n$. Since $(x_{nm}) \rightarrow x_n$, pick $l \geq M$ such that $|x_{nl} - x_n| < 1/n$.

Consider the sequence $(x_{nl}) \subset \mathbb{Q}^\omega$. By construction, $(a^{x_{nl}}) \sim (a^{x_n})$. Since $(x_{nl}) \sim (x_n)$ and $(x_n) \rightarrow x$, we have $(x_{nl}) \rightarrow x$, so $(a^{x_{nl}}) \rightarrow a^x$ by uniqueness of limits of rational sequences for exponentiation, so $(a^{x_n}) \rightarrow a^x$ as well.

4 Formal Definition of Exponential Function

Define $f_n : [a, b] \rightarrow \mathbb{R}$ by $f_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$. Let $M = \max(|a|, |b|)$. I claim that (f_n) converges in the Uniform metric ρ .

Observe that $\sum_{k=0}^{\infty} \frac{M^k}{k!}$ converges absolutely by the Ratio test, so the sequence of partial sums is Cauchy.

Let $\epsilon > 0$. Pick $N \in \mathbb{Z}^+$ such that $m > n \geq N \rightarrow \sum_{k=n+1}^m \frac{M^k}{k!} < \epsilon$. Then

$$\begin{aligned} \rho(f_m, f_n) &= \sup_{x \in [a, b]} \left| \sum_{k=0}^m \frac{x^k}{k!} - \sum_{k=0}^n \frac{x^k}{k!} \right| = \sup_{x \in [a, b]} \left| \sum_{k=n+1}^m \frac{x^k}{k!} \right| \\ &\leq \sup_{x \in [a, b]} \sum_{k=n+1}^m \frac{|x|^k}{k!} \leq \sum_{k=n+1}^m \sup_{x \in [a, b]} \frac{|x|^k}{k!} = \sum_{k=n+1}^m \frac{M^k}{k!} < \epsilon. \end{aligned}$$

Hence, (f_n) is Cauchy. Since $\mathcal{C}([a, b], \mathbb{R})$ is complete, (f_n) converges to an element $f \in \mathcal{C}([a, b], \mathbb{R})$. Rename this element $E : [a, b] \rightarrow \mathbb{R}$. Point-wise convergence implies $E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$.

Since the value $E(x)$ can be interpreted as the value of the series, $E(x)$ is global. Restrict E to $E'(a, b)$, take $(a, b) = (-n, n)$ and union over $n \in \mathbb{Z}^+$. Then we can define the continuous function $E : \mathbb{R} \rightarrow \mathbb{R}$ by $E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$.

5 Special Cases

Let $q : \mathbb{N} \rightarrow \mathbb{Q}$ be a bijection and define $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} 2^{-q^{-1}(x)} & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Observe that for $x \in \mathbb{Q}$, $g(x) > 0$, but we can pick sequence (x_n) of irrationals with $x_n \rightarrow x$, so $g(x_n) \rightarrow 0$, so g is discontinuous at x .

For $x \notin \mathbb{Q}$, $g(x) = 0$. Let (x_n) be any sequence converging to x . Suppose $\epsilon > 0$. By Archimedean principle, pick $N_1 \in \mathbb{Z}^+$ such that $1/N_1 < \epsilon$, so $2^{-N_1} < 1/N_1 < \epsilon$. Pick N_2 such that $n \geq N_2 \rightarrow x_n \neq q_0, q_1, q_2, \dots, q_{N_1}$, so $g(x_n) < 2^{-N_1} < \epsilon$. g is nonnegative, so $g(x_n) \rightarrow 0$, so g is continuous at x .

Define $f(x) = \sum_{r < x, r \in \mathbb{Q}} 2^{-q^{-1}(r)}$. Since $\sum_{n=0}^{\infty} 2^{-n} = 2$ is absolutely convergent, so is $\sum_{r \in \mathbb{Q}} 2^{-q^{-1}(r)}$, so f is well-defined.

(a). f is strictly increasing.

Suppose $x < y$. By density of $\mathbb{Q}$ in $\mathbb{R}$, pick $x < s < y$ with $s \in \mathbb{Q}$. Then

$$\begin{aligned} f(x) &= \sum_{r \in (-\infty, x) \cap \mathbb{Q}} 2^{-q^{-1}(r)} < \sum_{r \in (-\infty, x) \cap \mathbb{Q}} 2^{-q^{-1}(r)} + 2^{-q^{-1}(s)} \\ &\leq \sum_{r \in (-\infty, x) \cap \mathbb{Q}} 2^{-q^{-1}(r)} + \sum_{r \in [x, y) \cap \mathbb{Q}} 2^{-q^{-1}(r)} = \sum_{r \in (-\infty, y) \cap \mathbb{Q}} 2^{-q^{-1}(r)} \\ &= f(y). \end{aligned}$$

(b). f is discontinuous at rational points.

Let $x_0 \in \mathbb{Q}$. Pick any sequence (l_n) of points approaching x_0 from the left. Pick (r_n) approaching x_0 from the right. Then $f(l_n) < f(x_0)$ by part (a). Further, for any $y > x_0$,

$$\begin{aligned} f(y) &= \sum_{r \in (-\infty, y) \cap \mathbb{Q}} 2^{-q^{-1}(r)} = \sum_{r \in (-\infty, x_0) \cap \mathbb{Q}} 2^{-q^{-1}(r)} + 2^{-q^{-1}(x_0)} + \sum_{r \in (x_0, y) \cap \mathbb{Q}} 2^{-q^{-1}(r)} \\ &> f(x_0) + 2^{-q^{-1}(x_0)}. \end{aligned}$$

It follows that if $f(r_n)$ has a limit, it is at least $f(x_0) + 2^{-q^{-1}(x_0)}$, which is strictly greater than the limit of $f(l_n)$, if it exists.

(c). f is continuous at irrational points.

Let (x_n) converge to $x_0 \notin \mathbb{Q}$. Let $\epsilon > 0$. Suppose $x > x_0$. Pick $N \in \mathbb{Z}^+$ such that $2^{-N} < \epsilon$. Since $\mathbb{R}$ is T1 and $x_0 \neq q(0), q(1), \dots, q(N)$, pick $\delta > 0$ such

that $V_\delta(x_0)$ doesn't contain any $q(0), q(1), \dots, q(N)$. Take $x < x_0 + \delta$. Then $\sum_{r \in (x_0, x) \cap \mathbb{Q}} 2^{-q^{-1}(r)} \leq \sum_{k > N} 2^{-k} = 2^{-N} < \epsilon$. It follows that

$$\begin{aligned} f(x) - f(x_0) &= \sum_{r \in [x_0, x) \cap \mathbb{Q}} 2^{-q^{-1}(r)} \\ &= \sum_{r \in (x_0, x) \cap \mathbb{Q}} 2^{-q^{-1}(r)} < \epsilon. \end{aligned}$$

Since $f(x) > f(x_0)$, we have $x \in [x_0, x_0 + \delta) \rightarrow |f(x) - f(x_0)| < \epsilon$.

Now suppose $x < x_0$. Pick $\delta > 0$ such that $V_\delta(x_0)$ doesn't contain any $q(0), q(1), \dots, q(N)$. Take $x > x_0 - \delta$. Then

$$f(x) - f(x_0) = \sum_{r \in [x, x_0) \cap \mathbb{Q}} 2^{-q^{-1}(r)} \leq \sum_{r \in (x_0 - \delta, x_0) \cap \mathbb{Q}} 2^{-q^{-1}(r)} < \epsilon.$$

Hence, $f(V_\delta(x_0)) \subset V_\epsilon(f(x_0))$.

6 Differentiation

Let $f : X \rightarrow \mathbb{R}$ be a function. f is said to be differentiable if for all limit points $x_0 \in X$, the limit $\lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x) - f(x_0)}{x - x_0}$ exists.

Claim: differentiability implies continuity. Suppose $x_0 \in X$. If x_0 is isolated, there exists $\delta > 0$ such that $V_\delta(x_0) \cap X = \{x_0\}$. Then

$$\lim_{x \rightarrow x_0; x \in X} f(x) = \lim_{x \rightarrow x_0; x \in X \cap V_\delta(x_0)} f(x) = \lim_{x \rightarrow x_0; x \in X \cap V_\delta(x_0)} f(x_0) = f(x_0),$$

so f is continuous at x_0 . Else, x_0 is a limit point, so $x_0 \in \overline{X - x_0}$. By assumption,

$$\lim_{x \rightarrow x_0; x \in X - x_0} f(x) - f(x_0) = \lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \lim_{x \rightarrow x_0; x \in X - x_0} (x - x_0).$$

The right limit of the RHS evaluates to

$$\lim_{x \rightarrow x_0; x \in X - x_0} (x - x_0) = \lim_{x \rightarrow x_0; x \in X} (x - x_0) = \lim_{x \rightarrow x_0; x \in X} x - \lim_{x \rightarrow x_0; x \in X} x_0 = x_0 - x_0 = 0,$$

so the entire limit evaluates to 0, ie $\lim_{x \rightarrow x_0; x \in X - x_0} f(x) = f(x_0)$.

Suppose $f, g : X \rightarrow \mathbb{R}$ are differentiable. Then $f + g$ is differentiable, and at every limit point $x_0 \in X$, we have $(f + g)' = f' + g'$.

Note that trivially the limit points of X are the same with respect to f, g (property of X). We have:

$$\begin{aligned} \lim_{x \rightarrow x_0; x \in X - x_0} \frac{(f+g)(x) - (f+g)(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} + \frac{g(x) - g(x_0)}{x - x_0} \right) \\ &= \lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x) - f(x_0)}{x - x_0} + \lim_{x \rightarrow x_0; x \in X - x_0} \frac{g(x) - g(x_0)}{x - x_0}. \end{aligned}$$

Product rule.

$$\begin{aligned} \lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x)(g(x) - g(x_0)) + g(x_0)(f(x) - f(x_0))}{x - x_0} \\ &= \lim_{x \rightarrow x_0; x \in X - x_0} f(x) \cdot \frac{g(x) - g(x_0)}{x - x_0} + g(x_0) \cdot \lim_{x \rightarrow x_0; x \in X - x_0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= f(x_0)g'(x_0) + g(x_0)f'(x_0). \end{aligned}$$

I used that $\lim_{x \rightarrow x_0; x \in X - x_0} f(x) = f(x_0)$, proved earlier.

Chain Rule. Let $X, Y \subset \mathbb{R}$, $f : X \rightarrow Y, g : Y \rightarrow \mathbb{R}$, $x_0 \in X, y_0 \in Y$ st $y_0 = f(x_0)$, x_0, y_0 are limit points of X, Y , resp, and f, g are differentiable at x_0, y_0 , resp. Show that $gf : X \rightarrow \mathbb{R}$ is diff at x_0 , and has derivative $g'(y_0)f'(x_0)$.

Proof: Let $\epsilon > 0$. Since g is diff at y_0 , let $\delta_1 > 0$ st $y \in V_{\delta_1}(y_0) \cap Y \rightarrow |g(y) - (g(y_0) + (y - y_0)g'(y_0))| \leq \epsilon|y - y_0|$. Since f is diff at x_0 , f is cont at x_0 . Pick $\delta_2 > 0$ st $x \in V_{\delta_2}(x_0) \rightarrow f(x) \in V_{\delta_1}(y_0) \cap Y$.

Then for $x \in V_{\delta_2}(x_0)$, we have $|gf(x) - (g(y_0) + (f(x) - y_0)g'(y_0))| \leq \epsilon|f(x) - y_0|$. For $x \neq x_0$, we have

$$\left| \frac{gf(x) - gf(x_0) - (f(x) - f(x_0))g'(y_0)}{x - x_0} \right| \leq \epsilon \left| \frac{f(x) - f(x_0)}{x - x_0} \right|.$$

Pick $\delta_2 > \delta_3 > 0$ such that $\left| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right| \leq \epsilon$. Pick $\delta_4 > 0$ a constant independent of ϵ such that $\frac{f(x) - f(x_0)}{x - x_0}$ is bounded over $V_{\delta_4}(x_0) \cap X$ by C . Then for $x \in V_{\min(\delta_3, \delta_4)}(x_0) \cap X$, we have

$$\left| \frac{gf(x) - gf(x_0)}{x - x_0} - g'(y_0)f'(x_0) \right| \leq \epsilon|g'(y_0)| + C\epsilon = \epsilon(|g'(y_0)| + C).$$

Since $|g'(y_0)| + C$ is constant, $\lim_{x \rightarrow x_0; x \in X - x_0} \frac{gf(x) - gf(x_0)}{x - x_0} = g'(y_0)f'(x_0)$, as desired.

Let $\epsilon > 0$. Pick $q \in \mathbb{Q}$ such that $\alpha - \epsilon < q < \alpha$. Then for $x \in (0, \infty) - 1$,

we have $\frac{x^q-1}{x-1} \leq \frac{x^\alpha-1}{x-1}$. Since $\lim_{x \rightarrow 1; x \in (0, \infty) - 1} \frac{x^q-1}{x-1} = q$, pick $\delta > 0$ such that $x \in V_\delta(1) \cap ((0, \infty) - 1) \rightarrow \frac{x^q-1}{x-1} > q - (q - \alpha + \epsilon) = \alpha - \epsilon$. Then for $x \in V_\delta(1) \cap ((0, \infty) - 1)$, we have $\frac{x^\alpha-1}{x-1} > \alpha - \epsilon$. Similarly, we can pick $\delta' > 0$ such that $x \in V_{\delta'}(1) \cap ((0, \infty) - 1)$ implies $\frac{x^\alpha-1}{x-1} < \alpha + \epsilon$. Combining, we have

$$x \in V_{\min(\delta, \delta')}(1) \cap ((0, \infty) - 1) \rightarrow \alpha - \epsilon < \frac{x^\alpha - 1}{x - 1} < \alpha + \epsilon,$$

so $\lim_{x \rightarrow 1; x \in (0, \infty) - 1} \frac{x^\alpha - 1}{x - 1} = \alpha$.

It follows for all $x_0 > 0$, we have $\lim_{x \rightarrow x_0; x \in (0, \infty) - x_0} \frac{x^\alpha - x_0^\alpha}{x - x_0} = \lim_{h \rightarrow 1; h \in (0, \infty) - 1} \frac{x_0^\alpha (h^\alpha - 1)}{x_0(h - 1)} = x_0^{\alpha-1} \lim_{h \rightarrow 1; h \in (0, \infty) - 1} \frac{h^\alpha - 1}{h - 1} = \alpha x_0^{\alpha-1}$.

L'Hopital. Let $a < b$. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be diff on $[a, b]$. Suppose $f(a) = g(a) = 0$ and $g'(x) \neq 0$ for all $x \in (a, b]$. Then $g(x) > 0$ for all $x \in (a, b]$. Suppose further $\lim_{x \rightarrow a; x \in (a, b]} \frac{f'(x)}{g'(x)} = L$. Then $\lim_{x \rightarrow a; x \in (a, b]} \frac{f(x)}{g(x)} = L$.

Proof: Suppose FTSOC $g(x) = 0$ for some $x \in (a, b]$. Observe that g is diff on $[a, x]$, so by Rolle's $g'(c) = 0$ for some $c \in (a, x]$.

Since $\lim_{x \rightarrow a; x \in (a, b]} \frac{f'(x)}{g'(x)} = L$, pick $\delta > 0$ such that $x \in V_\delta(a) \cap (a, b] \rightarrow \frac{f'(x)}{g'(x)} \in V_\epsilon(L)$. Let $x \in V_\delta(a) \cap (a, b]$. Since f, g are diff on $[a, x]$, by Joint MVT, there exists $c \in (a, x)$ such that $(f(x) - f(a))g'(c) = (g(x) - g(a))f'(c) \rightarrow f(x)g'(c) = g(x)f'(c)$. Since $g, g' \neq 0$ on $(a, b]$, we have $\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}$ for $c \in V_\delta(a) \cap (a, b]$. It follows $\frac{f(x)}{g(x)} \in V_\epsilon(L)$, so $\lim_{x \rightarrow a; x \in (a, b]} \frac{f(x)}{g(x)} = L$, as desired.